PrismTone: Instant Persona and Communication Transformer

Vibeathon 2026 | Graphic Era Hill University, Bhimtal Campus | Problem 1: Artificial Intelligence

One rough draft, many voices, and proof of exactly what changed.

1. Name Suggestions

The name should be short, easy to say on stage, and hint at "one input, many outputs."

Name	Why it works
PrismTone (recommended)	A prism splits one beam into many colours, just as the tool splits one draft into many voices. It is easy to remember and easy to brand.
VoiceForge	Suggests crafting voices on demand. Strong and techy.
ToneShift	Describes the action directly. Very clear, but a common name.
Chameleon	One message that adapts to every audience. Playful and memorable.
Echo Persona	Your message, echoed in many voices.
SayItRight	Plain-language, instantly understood by non-technical judges.
VibeShift	Ties into the Vibeathon theme.

Suggested tagline: *Write once. Sound right to everyone.*

Before finalizing, search the name online and in app stores to check that it is not already taken.

2. Problem and Vision

Problem

People write rough drafts under time pressure, and the same message must sound different for a CEO, a teammate, a legal team or a younger audience. Rewriting by hand is slow, and tone mistakes cause real damage: blame, confusion, missed commitments or legal risk.

Vision

PrismTone turns raw text or bullet points into audience-ready communication in seconds, and, unlike a plain rewriter, **shows exactly how the rewrite differs from the original** and **proves that the facts were preserved**.

Target users

- Students and early-career professionals who are unsure about tone in emails.
- Managers giving feedback or reporting upward.
- Support, HR and operations staff writing notices.
- Founders and team leads who communicate across several audiences daily.

3. Core Features (Must Build)

1. **Draft input:** a text box for plain text or bullet points, with a character counter and sample drafts for quick demos.
2. **Four default personas, generated in parallel:**
   - **Executive C-Level:** 2 to 3 crisp lines, outcome first, with the ask and deadline clear.
   - **Supportive Peer:** warm, constructive, appreciative, with the critique framed as growth.
   - **Legal/Compliance:** neutral, precise, factual, no emotive language or blame, with commitments and dates stated exactly.
   - **Gen-Z Hype:** energetic, casual and short, with light slang (kept professional enough to demo).
3. **Persona cards:** each card has the rewritten text, a copy button, and a regenerate button.
4. **Change visualization (the key differentiator):**
   - Word-level diff between the original and each rewrite (additions, deletions, kept text).

    - Change metrics per persona: length change (%), reading grade, formality, warmth, directness.
    - A radar chart comparing the original with each persona.
    - A single **"How much did it change?"** score from 0 to 100 (for example, a blend of word-level edit distance and tone shift).
5. **Fast, streamed output** so users see results appear within seconds.

4. Advanced Features (Differentiators)

4.1 Trust and safety

- **Fact Guard:** the model extracts numbers, dates, names, amounts and commitments from the draft, and the app checks that each one appears in every rewrite. It flags anything **dropped** or **invented**, shown as a green/amber/red checklist.
- **Meaning-preserved score:** a similarity score between the original and each rewrite, so users can see whether the intent survived.
- **Risk flags** (especially for Legal and Executive): absolute promises ("guarantee", "always"), blame language, vague deadlines ("soon"), unsupported claims, and ambiguous commitments.
- **Privacy mode:** automatically mask names, emails and phone numbers before sending text to the AI model, then restore them in the output. No drafts stored by default.

4.2 Personalization

- **Custom persona sliders:** formality, warmth, brevity, directness and humour. Each slider changes the prompt, and the radar chart moves live.
- **Create your own persona:** for example "Investor update", "Parent-teacher note" or "Customer apology", saved for reuse.
- **Voice matching:** paste a sample of your own writing and PrismTone imitates your style so the output still sounds like you.
- **Audience presets for students:** professor email, internship application, club announcement, placement HR reply.

4.3 Explainability

- **"Why this changed" notes:** short annotations on key edits (for example, "Moved the deadline to the first line because executives scan for the ask").
- **Before/after tone meter:** a simple bar showing where the original tone sits and where the rewrite sits.

4.4 Output and workflow

- **Channel formats:** email (with subject line), Slack, WhatsApp, LinkedIn, SMS. Each shapes length and structure.
- **Length control:** one-liner, short, standard.
- **Multi-language output:** English, Hindi and Hinglish (a strong local advantage).
- **Reverse mode (Tone Decoder):** paste an incoming message and PrismTone explains its tone, hidden urgency and suggested reply styles.
- **Compare mode:** view any two personas side by side with their diff.
- **History and favourites:** local storage, with an export option.
- **Voice input and read-aloud:** speak a rough draft, then hear the polished version.
- **Shareable link or PDF export:** share a comparison with a teammate.
- **Browser extension or Gmail-style inline widget (stretch goal):** use it where people already write.

5. Suggested Technical Design

Stack

- **Front end:** React or Next.js, Tailwind CSS, Recharts (radar chart), diff-match-patch (word diff).
- **AI:** an LLM API (Claude or Gemini) called from a small server route so the API key stays hidden.
- **Storage:** browser local storage for history and custom personas (no database needed in six hours).
- **Hosting:** Vercel or Netlify.

Data flow

1. User enters a draft and picks personas and sliders.
2. The privacy layer masks sensitive details.
3. One request per persona is sent in parallel, asking for **structured JSON**.
4. The app parses the JSON, then runs the diff, the fact check and the metrics, and renders the cards.

Example JSON returned per persona

```
{
  "persona": "executive",
  "rewrite": "...",
  "tone_scores": {"formality": 8, "warmth": 3, "directness": 9},
  "facts_found": ["Friday deadline", "server outage"],
  "risk_flags": ["blame language removed"],
  "change_notes": ["Led with the deadline", "Removed personal blame"]
}
```

Scoring ideas

- **Change score:** word-level edit distance normalised to 0 to 100, blended with the absolute tone-score shift.
- **Fact Guard:** run an extraction pass on the original, then check each item against each rewrite (exact match first, model check for paraphrases).
- **Reading grade:** Flesch-Kincaid computed in the browser.

Reliability tips

- Validate the JSON and retry once on failure.
- Keep **offline sample data** so the demo works even if Wi-Fi or the API fails.
- Set a timeout per persona so one slow request does not block the rest.

6. Market Competitors

Features change often. Re-check each tool before presenting.

Competitor	Strengths	Weaknesses	Opening for PrismTone
Grammarly	Trusted, polished, tone detection, works everywhere	Generic tone options, no side-by-side personas, advanced features paywalled, little explanation of changes	Show many audiences at once with measurable change
Wordtune	Fast tone and length toggles, good sentence-level rewrites	Works sentence by sentence, no change metrics, no legal or executive presets	Whole-message audience targeting with a change score

Competitor	Strengths	Weaknesses	Opening for PrismTone
QuillBot	Several paraphrase modes, highlights changed words	Paraphrases rather than re-targeting for an audience, can drift from the facts	Fact Guard and audience-specific logic
Goblin.tools (Formalizer)	Free and simple	One direction only, no analysis	Multi-persona output with analytics
ChatGPT, Claude, Gemini	Very flexible, strong writing	One result per prompt, no diff or scoring, users must write good prompts	A purpose-built interface with zero prompt writing and built-in verification
Gmail Help me write, Outlook Copilot	Built into the inbox	Locked to one platform, little visibility of what changed	Platform-independent and transparent
Lavender, Textio	Email coaching and bias-aware writing for sales and hiring	Narrow use cases, paid business tools	Broader everyday communication with a free demo

Common gaps across competitors

1. They give **one** rewrite, not several audiences side by side.
2. They rarely **quantify** how much the message changed.
3. They do not **verify that facts survived**.
4. They rarely **explain** why edits were made.
5. Privacy handling is often unclear.

7. How to Overtake the Competition

Strategy	What to do
Own transparency	Make the diff, change score and "why this changed" notes the star of the demo.
Build trust	Fact Guard and risk flags show that the rewrite did not lose or invent anything. Competitors rarely offer this.

Strategy	What to do
Show all voices at once	Parallel persona cards plus compare mode reduce the effort of trying personas one at a time.
Zero prompt skill needed	Users choose sliders and personas rather than writing prompts.
Privacy first	Mask personal data before sending and store nothing by default.
Local advantage	Hindi and Hinglish support, plus student scenarios such as professor emails and placement messages.
Learning value	Explain tone choices so users improve their own writing over time.
Speed and simplicity	A clean UI with first results in under 5 seconds, no sign-up needed.

Positioning statement

> For anyone who writes under pressure, PrismTone rewrites one draft for every audience and shows exactly what changed and what stayed true. Unlike general rewriters, it measures the change and checks the facts.

8. Six-Hour Build Plan

Hour	Goal
1	Finalize name and stack, set up the project, write and test the structured JSON prompt.
2	Build the input UI and the parallel persona cards.
3	Connect streaming, copy and regenerate, and error handling.
4	Add the word diff, tone metrics and radar chart.
5	Add Fact Guard, risk flags, sliders and the privacy masking.
6	Polish the UI, add fallback sample data, deploy, and rehearse the demo.

Priority rule: finish everything in Section 3 first, then add features from Section 4 in this order: Fact Guard, sliders, why-notes, risk flags, channel formats, then extras.

9. Demo Script (3 to 4 minutes)

1. **Hook:** "Everyone has sent a message that landed badly."
2. **Input a messy draft:** *"Server went down, it's Raj's fault, fix by Friday."*
3. **Show the four personas appear** side by side.
4. **Open the diff and radar chart:** "Here is exactly how much each version changed."
5. **Show Fact Guard:** "Friday and the outage are preserved in every version, and blame was removed from the Legal and Executive versions."
6. **Move a slider live** and watch the rewrite and chart update.
7. **Mention privacy** (names masked before reaching the AI) and Hinglish support.
8. **Close:** "PrismTone doesn't just rewrite your message. It shows you what changed and proves nothing was lost."

10. Judging Checklist

- Works end to end with a live demo.
- Clear differentiator (change visualization and Fact Guard).
- Handles edge cases (empty input, very long text, API failure).
- Clean, responsive interface.

- Clear explanation of who benefits and why it beats existing tools.
- Privacy and responsible AI considered.

11. Possible Future Scope

- Browser extension and Gmail or Slack plugin.
- Team style guides (company voice presets).
- Analytics on personal communication habits.
- Accessibility modes (plain language, dyslexia-friendly).
- An API for other apps to embed persona rewriting.